CIP-B101

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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Submitted By

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Course Information

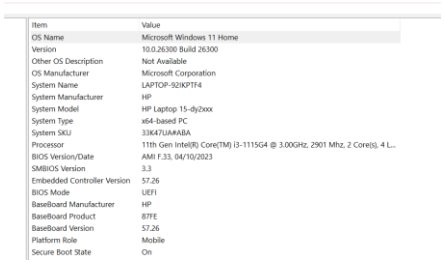
Item	Details
Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Title	PC System Fundamentals, Disk Sectors and Partition Analysis
Student Name	Sharon Naa Adjeley Adjetey Sowah
Student id	C11/26/PCDFIR/17026
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Tools used	Windows 11 pc, Calculator, Screenshot tool
Instructor	Basiru Aliyu
Institution	International Cybersecurity and Digital Forensics Academy (ICDFA)

Introduction

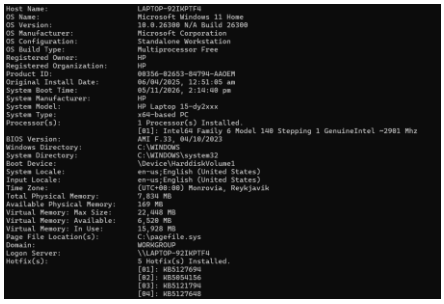
This laboratory exercise was conducted to develop a foundational understanding of computer systems and their relevance to digital forensic investigations. The purpose of the lab was to examine key system components, identify potential sources of digital evidence, and understand how data is stored and managed on a computer system. Practical activities included collecting system information, inspecting disk partitions, calculating partition sizes from sector data, analyzing the PC boot process, and comparing MBR and GPT partitioning schemes. The lab also involved identifying file systems used on the computer and explaining their forensic significance. Throughout the exercise, only safe read-only methods were used to observe and document system information without altering any data, following forensic best practices. These skills provide a foundation for conducting systematic digital evidence collection and analysis in future investigations.

Part A - System Identification and Evidence Layer Mapping

A1. Capture basic system specification



Item	Value
OS Name	Microsoft Windows 11 Home
Version	10.0.26300 Build 26300
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	LAPTOP-92IKPTF4
System Manufacturer	HP
System Model	HP Laptop 15-dy2xxx
System Type	x64-based PC
System SKU	33K47UA#ABA
Processor	11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz, 2901 Mhz, 2 Core(s), 4 L...
BIOS Version/Date	AMI F.33, 04/10/2023
SMBIOS Version	3.3
Embedded Controller Version	57.26
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	87FE
BaseBoard Version	57.26
Platform Role	Mobile
Secure Boot State	On



```
Host Name:                 LAPTOP-92IKPTF4
OS Name:                   Microsoft Windows 11 Home
OS Version:                10.0.26300 N/A Build 26300
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          HP
Registered Organization:   HP
Product ID:                00356-02653-84794-AAOEM
Original Install Date:     06/04/2025, 12:51:05 am
System Boot Time:          05/11/2026, 2:16:48 pm
System Manufacturer:       HP
System Model:              HP Laptop 15-dy2xxx
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 140 Stepping 1 GenuineIntel ~2981 Mhz
BIOS Version:              AMI F.33, 04/10/2023
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+00:00) Monrovia, Reykjavik
Total Physical Memory:     7,834 MB
Available Physical Memory: 169 MB
Virtual Memory: Max Size:  22,448 MB
Virtual Memory: Available: 6,320 MB
Virtual Memory: In Use:    15,928 MB
Page File Location(s):     C:\pageFile.sys
Domain:                    WORKGROUP
Logon Server:              \\LAPTOP-92IKPTF4
Hotfix(s):                 5 Hotfix(s) Installed.
                           [01]: KB5127694
                           [02]: KB5054156
                           [03]: KB5121796
                           [04]: KB5127648
```

A2. Capture storage information

```
PS C:\Users\sharo> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                             Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      NVMe SK hy... ACE4_2E00_1A80_3DC2_2EE4_AC00... Healthy              Online                  238.47 GB GPT
```

```
   DiskPath: \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 237.14 GB Basic
4                            254915117056                                1.06 GB Recovery
```

```
PS C:\Users\sharo> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C           Windows      NTFS           Fixed     Healthy      OK                    37.15 GB 237.14 GB
                         NTFS           Fixed     Healthy      OK                   114.36 MB   1.06 GB
```

A3. Map evidence sources to system layers

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	User accounts, login records, user profiles, documents, preferences, and personal activity records.	Recent files, browser history, shell history	Helps identify which user account was involved in activities and provides information about user actions and behavior.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
Application layer	Browser history, downloaded files, chat records, application logs, and application databases.	Application folders and databases	Helps investigators identify user activities, communication, browsing history, and application usage.
OS layer	System event logs, security logs, file access records, timestamps, installed programs, and system configuration.	Windows logs, registry, Linux /var/log	Helps investigators reconstruct system activities, identify login events, and establish when certain activities occurred.
Disk layer	Files, deleted files, file metadata, partitions, file systems, and stored application data.	File system and unallocated space	Provides persistent evidence of files created, modified, deleted, or stored on the computer.
Memory/process layer	Running processes, active network connections, temporary data, encryption keys, and currently active application information.	RAM capture, process list	Helps investigators identify activities occurring while the computer is running and recover volatile evidence that may disappear when the computer is shut down.

Digital forensic evidence may exist in more than one system layer because computer activities involve applications, users, the operating system, disk storage, and memory working together. For example, when a user downloads a file using a web browser, evidence may be generated in the application layer through browser history, in the user layer through the user's activity, in the OS layer through file records and timestamps, on the disk through the downloaded file, and in memory through temporary data. Therefore, investigators examine multiple layers to obtain a complete picture of user activities, correlate evidence, and verify findings.

Part B - Program, Process and Thread Investigation

B1. Observe running processes

Process name	PID	Likely Purpose
winlogon.exe	808	Manages Windows user sign-in and sign-out processes and the user session.
services.exe	1048	Manages Windows services and starts or controls background services.
dwm.exe	1520	Desktop Window Manager, responsible for displaying and

		managing desktop windows and visual effects.

```
C:\Users\sharo>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        860 K
Secure System                   92 Services                   0     40,904 K
Registry                       136 Services                   0      9,804 K
smss.exe                       608 Services                   0          N/A
csrss.exe                      948 Services                   0      2,688 K
wininit.exe                    556 Services                   0        732 K
csrss.exe                      724 Console                    1      2,680 K
winlogon.exe                   808 Console                    1      1,256 K
services.exe                  1048 Services                   0      5,676 K
LsaIso.exe                    1076 Services                   0      1,020 K
lsass.exe                     1084 Services                   0     12,436 K
svchost.exe                   1212 Services                   0     19,220 K
fontdrvhost.exe               1228 Services                   0        244 K
fontdrvhost.exe               1220 Console                    1      1,280 K
WUDFHost.exe                  1344 Services                   0        804 K
svchost.exe                   1380 Services                   0     12,424 K
svchost.exe                   1424 Services                   0      2,892 K
dwm.exe                       1520 Console                    1     69,308 K
svchost.exe                   1600 Services                   0      1,632 K
svchost.exe                   1608 Services                   0        624 K
svchost.exe                   1736 Services                   0      5,228 K
svchost.exe                   1752 Services                   0      1,944 K
WUDFHost.exe                  1844 Services                   0      5,336 K
svchost.exe                   1852 Services                   0      1,944 K
svchost.exe                   1860 Services                   0      1,924 K
svchost.exe                   1880 Services                   0      2,396 K
svchost.exe                   1904 Services                   0      5,684 K
svchost.exe                   2012 Services                   0      4,396 K
svchost.exe                   1272 Services                   0      1,412 K
IntelCpHDCPSvc.exe            1744 Services                   0        504 K
svchost.exe                   2052 Services                   0      8,124 K
svchost.exe                   2064 Services                   0      3,248 K
svchost.exe                   2120 Services                   0        984 K
```

B2. Explain the difference between Program, Process and Thread

A program is a set of instructions stored on a computer that tells the computer what to do, while a process is a program that is currently running and using system resources such as memory and CPU time. A thread is the smallest unit of execution within a process, and a process can contain multiple threads that perform different tasks at the same time. For example, Google Chrome is a program installed on the computer, and when it is opened, Windows creates one or more processes to run it. Each process has a unique Process Identification Number (PID), which helps the operating system identify and manage it. Within these processes, threads can perform different activities, such as loading webpages, responding to user actions, and running background tasks. Understanding the difference between programs, processes, and threads is important in digital forensics because it helps investigators identify running applications, examine their activities, monitor system resource usage, and detect suspicious processes or threads that may be associated with malware. During live forensic triage, process information can also help investigators identify activities occurring on the computer before the system is shut down and volatile evidence is lost.

Explain why process information may be useful during malware analysis or live forensic triage

Process information is useful during malware analysis and live forensic triage because it helps investigators identify programs currently running on a computer and determine whether any suspicious activity is occurring. By examining process names, PIDs, CPU usage, and memory usage, investigators can identify unusual processes, unexpected resource consumption, or programs that may be running without the user's knowledge. Process information can also help investigators identify malware that is actively executing, determine which processes may require further examination, and understand how suspicious programs interact with other applications. During live forensic triage, this information is particularly valuable because running processes may contain evidence that could disappear when the computer is shut down. Therefore, documenting process information helps investigators understand the computer's current state and identify possible evidence of malicious activity.

Part C - Disk Sector and Disk Size Calculation

C2. Calculate your own disk or partition size

Screenshots

```
PS C:\Users\sharo> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                  Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      NVMe SK hy... ACE4_2E00_1A80_3DC2_2EE4_AC00... Healthy              Online                  238.47 GB GPT
```

```
   DiskPath: \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 237.14 GB Basic
4                            254915117056                                1.06 GB Recovery
```

```
PS C:\Users\sharo> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C           Windows      NTFS           Fixed     Healthy      OK                    37.15 GB 237.14 GB
                         NTFS           Fixed     Healthy      OK                   114.36 MB   1.06 GB
```

```
PS C:\Users\sharo> Get-Partition | Select-Object PartitionNumber,DriveLetter
,@{Name="StartSector";Expression={$_.Offset/512}},@{Name="SectorCount";Expre
ssion={$_.Size/512}},@{Name="SizeGB";Expression={$_.Size/1GB}}


PartitionNumber : 1
DriveLetter     :
StartSector     : 2048
SectorCount     : 532480
SizeGB          : 0.25390625

PartitionNumber : 2
DriveLetter     :
StartSector     : 534528
SectorCount     : 32768
SizeGB          : 0.015625

PartitionNumber : 3
DriveLetter     : C
StartSector     : 567296
SectorCount     : 497313792
SizeGB          : 237.1376953125

PartitionNumber : 4
DriveLetter     :
StartSector     : 497881088
SectorCount     : 2222080
SizeGB          : 1.0595703125
```

Calculations

Total Sectors = Partition 1 + Partition 2 + Partition 3 + Partition 4

= 532,480 + 32,768 + 497,313,792 + 2,222,080

= 500,101,120 sectors

Total Bytes = Total sectors × Bytes per sector (512)

= 500,101,120 x 512

= 256,051,73,440 bytes

Kilobytes = Total bytes / 1024

= 256,051,73,440 / 1024

= 250,050,560 KB

Megabytes = Total Kilobytes / 1024

= 250,050,560 / 1024

= 244,190 MB

Gigibytes = Total Megabytes / 1024

= 244,190 / 1024

= 238.46696875 GB

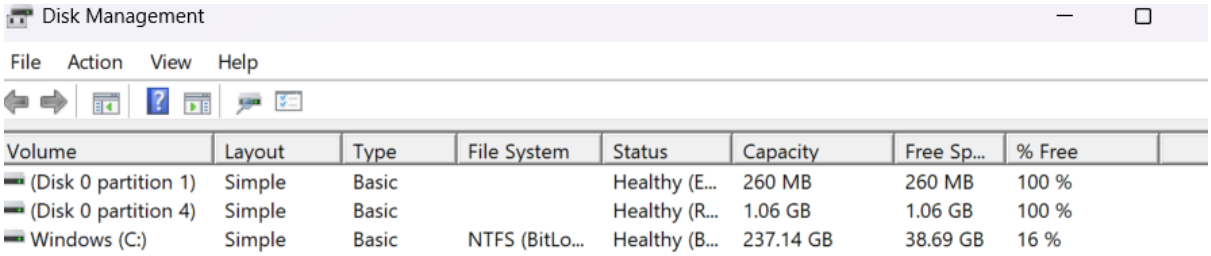
= 283.47 GB

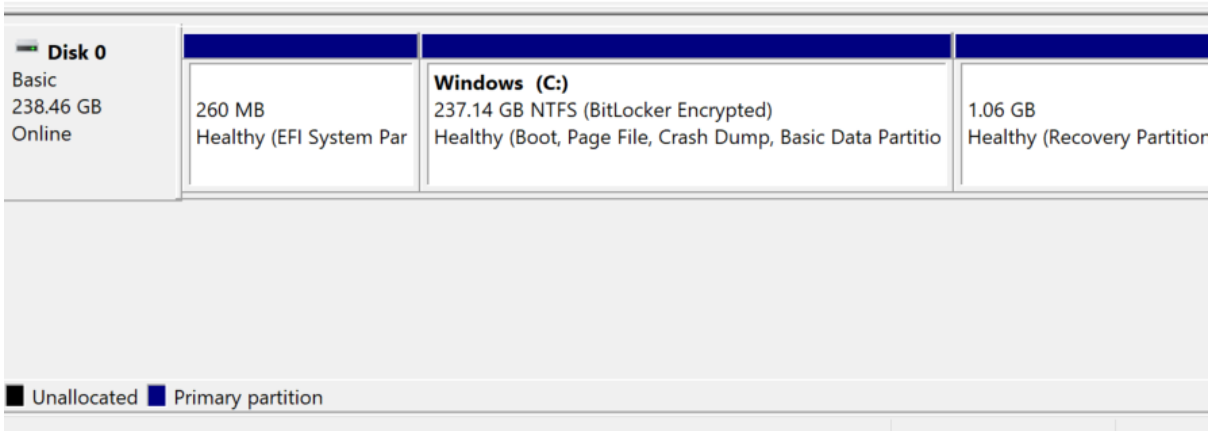
Table

Item	Formula / Value	Answer
Total sectors	532,480 + 32,768 + 497,313,792 + 2,222,080	500,101,120 sectors
Total bytes	500,101,120 × 512	256,051,773,440 bytes
Kilobytes	256,051,773,440 ÷ 1024	250,050,560 KB
Megabytes	250,050,560 ÷ 1024	244,190 MB
Gigabytes	244,190 ÷ 1024	238.47 GB

Part D - Safe Partition Table Inspection

D1. Windows method - Disk Management, PowerShell and DiskPart





Get Disk

```
PS C:\Users\sharo> Get-Disk

Number Friendly Name Serial Number                    HealthStatus         OperationalStatus      Total Size Partition
                                                                                                  Style
------ ------------- -------------                    ------------         -----------------      ---------- ----------
0      NVMe SK hy... ACE4_2E00_1A80_3DC2_2EE4_AC00... Healthy              Online                  238.47 GB GPT
```

Get Partition

```
   DiskPath: \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 237.14 GB Basic
4                            254915117056                                1.06 GB Recovery
```

Get Volume

```
PS C:\Users\sharo> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
C           Windows      NTFS           Fixed     Healthy      OK                     37.15 GB 237.14 GB
                         NTFS           Fixed     Healthy      OK                    114.36 MB   1.06 GB
```

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

Power On

↓

POST (Power-On Self-Test)

↓

BIOS/UEFI Firmware Loads

↓

CMOS / Firmware Settings Read

↓

Locate Boot Device

↓

MBR / GPT / EFI Boot Record

↓

Boot Loader / Boot Manager

↓

Kernel Loading

↓

Operating System Takes Control

- **Power-On**
  - The user presses the power button and electrical power is supplied to the computer's hardware.
- **POST (Power-On Self-Test)**
  - The system checks essential hardware such as the CPU, RAM, keyboard, and storage devices to make sure they are functioning correctly.
- **BIOS/UEFI Firmware**
  - The firmware initializes hardware and determines which device should be used to boot the system. Modern computers typically use UEFI while older systems use BIOS.
- **CMOS/Firmware Settings**
  - The boot order and hardware configuration settings stored in CMOS (for BIOS systems) or firmware settings (for UEFI systems) are applied.
- **MBR/GPT or EFI Boot Records**
  - BIOS systems look for the **Master Boot Record (MBR)** on the disk.
  - UEFI systems use the **EFI System Partition (ESP)** and GPT partition information to locate boot files.
- **Boot Loader/Boot Manager**
  - The boot manager loads the selected operating system and prepares the system for kernel execution. Examples include Windows Boot Manager and GRUB.
- **Kernel Loading**
  - The operating system kernel is loaded into memory and initializes system services, drivers, and hardware resources.
- **Operating System Control**
  - The operating system takes full control of the computer and presents the login screen or desktop. User activity and log creation begin at this stage.

My system uses **UEFI boot mode**. This was verified using **System Information (msinfo32)**, where the BIOS Mode field displayed **UEFI**.

BIOS Mode UEFI

Part F - File System Identification and Interpretation

Two identified filed system and the volumes used on my pc are :

 The **C:** volume, labeled **Windows**, uses the **NTFS (New Technology File System)** file system and has a total size of **237.14 GB**.
 A second volume with a size of **1.06 GB** also uses **NTFS**. This is likely a system or recovery partition used by Windows for recovery and maintenance purposes.

Importance of file systems in Digital Forensics

A file system determines how files are named, stored, organized, and retrieved from a storage device. According to the lecture slides, a file system defines how files are stored and retrieved from storage devices.In digital forensics, understanding the file system is important because investigators rely on file system structures and metadata to:

- Locate files and folders.
- Recover deleted data.
- Examine user activity.
- Analyze timestamps.
- Validate digital evidence.

Different file systems store metadata differently, making file system knowledge essential during forensic investigations

Part G - Final Practical Assignment

Tasks

1. IDENTIFY PARTITION 1

```
PS C:\Users\sharo> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 237.14 GB Basic
4                            254915117056                                1.06 GB Recovery
```

2. Record Partition 1 sector count

```
PS C:\Users\sharo> Get-Partition -DiskNumber 0 -PartitionNumber 1 | Select-Object PartitionNumber,Offset,Size,@{Name="StartSector";Expression={$_.Offset/512
}},@{Name="SectorCount";Expression={$_.Size/512}}


PartitionNumber : 1
Offset          : 1048576
Size            : 272629760
StartSector     : 2048
SectorCount     : 532480
```

Calculation of end sector:

End Sector=Start Sector +Sector Count - 1

=2048+532480-1

=534527

3. Calculate Partition 1 size

The partition table shows that Partition 1 contains 532,480 sectors. Each sector contains 512 bytes.

Calculating the size in bytes

Bytes= Sector Count * Sector Size

= 532480 * 512

= 272629760 bytes

Convert bytes to GB

GB = Bytes / 1024^3

=272629760 / 1073741824

= 0.254 GB

4. Identify Partition 2

```
PS C:\Users\sharo> Get-Partition -DiskNumber 0 -PartitionNumber 2 | Format-List *


OperationalStatus     : Online
Type                  : Reserved
DiskPath              : \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
ObjectId              : {1}\\LAPTOP-92IKPTF4\root/Microsoft/Windows/Storage/Providers_v2\WSP_Partition.ObjectId="{361e19cd-16a5-11ec-ad91-806e6f6e6963}:PR:
                        {00000000-0000-0000-0000-501000000000}\\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a
                        0c91efb8b}"
PassThroughClass      :
PassThroughIds        :
PassThroughNamespace  :
PassThroughServer     :
UniqueId              : {00000000-0000-0000-0000-501000000000}eui.ACE42E001A803DC2
AccessPaths           :
DiskId                : \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc711_h#4&2206bd65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
DiskNumber            : 0
DriveLetter           :
GptType               : {e3c9e316-0b5c-4db8-817d-f92df00215ae}
Guid                  : {71c7c47b-f58e-4354-895b-e3bfdb4d28d9}
IsActive              : False
IsBoot                : False
IsDAX                 :
IsHidden              : True
IsOffline             : False
IsReadOnly            :
IsShadowCopy          :
IsSystem              : False
MbrType               :
NoDefaultDriveLetter  :
Offset                : 273678336
PartitionNumber       : 2
Size                  : 16777216
TransitionState       : 1
PSComputerName        :
CimClass              : ROOT/Microsoft/Windows/Storage:MSFT_Partition
CimInstanceProperties : {ObjectId, PassThroughClass, PassThroughIds, PassThroughNamespace...}
CimSystemProperties   : Microsoft.Management.Infrastructure.CimSystemProperties
```

5. Record Partition 2 sector count

```
PS C:\Users\sharo> Get-Partition -DiskNumber 0 -PartitionNumber 2 | Select-Object PartitionNumber,Offset,Size,@{Name="StartSector";Expression={$_.Offset/512
}},@{Name="SectorCount";Expression={$_.Size/512}}


PartitionNumber : 2
Offset          : 273678336
Size            : 16777216
StartSector     : 534528
SectorCount     : 32768
```

Calculation of end sector:

End Sector =534528+32768-1
= 567295

6. Calculate Partition 2 size

The partition table shows that Partition 2 contains 32,768 sectors. Each sector contains 512 bytes.

Step 1: Calculate the size in bytes

Bytes = Sector Count * Sector Size

=32768 * 512

= 16777216 bytes

Step 2: Convert bytes to GB

GB = Bytes / 1024^3

= 16777216 / 1073741824

=0.015625 GB

7. **Compare with OS-reported size**

```
PS C:\Users\sharo> Get-Partition | Select-Object PartitionNumber,DriveLetter,Size

PartitionNumber DriveLetter         Size
--------------- -----------         ----
              1                 272629760
              2                  16777216
              3           C 254624661504
              4                1137704960
```

```
PS C:\Users\sharo> Get-Volume | Select-Object DriveLetter,FileSystem,Size,SizeRemaining

DriveLetter FileSystem         Size SizeRemaining
----------- ----------         ---- -------------
          C NTFS       254624657408   41978339328
            NTFS         1137700864     119914496
```

Partition	Partition table calculated size	Windows-reported size
Partition 1	0.254 GB	Approximately 260 MB
Partition 2	0.015 GB	Approximately 16 MB

8. Explain difference if any

The calculated partition sizes may differ slightly from the sizes displayed by Windows because Windows commonly displays partition sizes in MB or GB using binary units, while some tools use decimal units. In binary measurement, 1 GiB equals 1,073,741,824 bytes, while 1 GB equals 1,000,000,000 bytes. Windows may also round displayed values, which can cause small differences between the exact partition-table calculation and the displayed size. The partition table calculation uses the exact sector count multiplied by 512 bytes, while the operating system display may use rounded units.

Conclusion

This laboratory exercise provided practical experience in examining fundamental computer system concepts from a digital forensics perspective. Through the lab, I learned how computer system layers generate different types of digital evidence and how user activities can be traced through applications, the operating system, storage devices, and memory. I also gained experience identifying system specifications, inspecting disk partitions, and calculating disk and partition sizes using sector information. Additionally, I learned how the PC boot process works, including the roles of BIOS/UEFI, MBR/GPT, boot loaders, and the operating system. The lab further improved my understanding of common file systems such as NTFS and their forensic significance.

These concepts are important in digital forensics because investigators depend on system information, partition structures, boot records, file systems, and metadata to locate, preserve, analyze, and recover digital evidence. Understanding how data is stored and how a computer operates enables forensic analysts to conduct investigations accurately while maintaining the integrity of evidence. Overall, this lab established a strong foundation for more advanced forensic analysis and digital investigations